Automatic lathe

Automatic lathes are operated with complete automatic control. They are high speed, mass production lathe. An operator can took after more than are automatic lathe at a time.

Special purpose lathe

Special purpose lathes are used for special purposes and for jobs, which cannot be accommodated and conveniently machined on a standard lathe. Wheel lathe, 'T' lathe duplicating lathe are some examples of special purpose lathe.

1.13 Differences between a turret lathe and a capstan lathe

S.No	Turret lathe	Capstan lathe
1.	Turret tool head is directly fitted on the saddle and both of them appear like one unit.	Turret head is mounted on a slide called ram which is mounted on the saddle.
2.	Saddle is moved to provide feed to the tool.	To provide feed to the tool, saddle is locked at a particular point and the ram is moved.
3.	It is difficult to move the saddle for feed.	It is easy to move the ram for feed.
4.	As the saddle can be moved along the entire length of the bed. It is suitable for longer workpieces.	As the movement of the ram is limited, it is suitable for machining shorter workpieces only.
5.	To index the turret tool head, a clamping lever is released and the turret is rotated manually.	When the handwheel for the ram is reversed, the turret tool head is indexed automatically.
6.	Limit dogs are used to control the distance of tool movement.	To control the distance of tool movement, feed stop screws are provided at the rear side of the turret.
7.	Some turret lathes have the facility of moving the turret at right angles to the lathe axis.	No such facility.
8.	Heavy and sturdy.	Lighter in construction.
9.	Suitable for machining heavy and large workpieces.	Only small and light workpieces are machined.
10.	Machining can be done by providing more depth of cut and feed.	Only limited amount of feed and depth of cut are provided for machining.

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